

The 1/3–2/3 Conjecture and the #P-Completeness of Counting Linear Extensions

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Abstract

The 1/3–2/3 Conjecture is a central open problem in the study of finite partially ordered sets. It asserts that every finite poset that is not a chain has a pair of elements x, y such that the proportion of linear extensions in which x precedes y lies between 1/3 and 2/3. In this write up we introduce the conjecture, explain why the constant 1/3 is the natural threshold for this conjecture, and then turn to the exact counting problem that arises from its definition. Since these probabilities are expressed in terms of numbers of linear extensions, a relevant computational question is the problem of counting linear extensions. A theorem of Brightwell and Winkler shows that this problem is #P-complete.

1 Introduction

Let P be a finite partially ordered set. A linear extension of P is a total ordering of the elements of P that respects the given partial order. If P is already totally ordered, then there is only one possible linear extension. If P contains many incomparable pairs, then several different total orders may occur. The 1/3–2/3 Conjecture is concerned with the relative order of two elements across all linear extensions of a finite poset. Given two distinct elements $x, y \in P$, we can ask in what proportion of linear extensions x appears before y . The conjecture asserts that if P is not a chain, then there should always be a pair for which the proportion of linear extensions placing x before y lies between 1/3 and 2/3 [3].

2 Linear extensions and balance

Definition 2.1. Let P be a finite poset. A linear extension of P is a bijection $\ell : P \rightarrow \{1, 2, \dots, |P|\}$ such that whenever $x <_P y$, one has $\ell(x) < \ell(y)$.

Equivalently, a linear extension is a total ordering of the ground set of P that extends the given partial order. Let $\mathcal{L}(P)$ denote the set of linear extensions of P , and let $e(P) = |\mathcal{L}(P)|$. If $x, y \in P$ are distinct, define

$$\mathbb{P}_P(x < y) = \frac{\#\{\ell \in \mathcal{L}(P) : \ell(x) < \ell(y)\}}{e(P)}.$$

This is the proportion of linear extensions in which x appears before y . It is clear that if x and y are comparable, then $\mathbb{P}_P(x < y)$ is either 0 or 1. The interesting case is when x and y are incomparable.

Definition 2.2. Let $0 \leq \alpha \leq \frac{1}{2}$. A pair (x, y) of distinct elements of P is called α -balanced if

$$\alpha \leq \mathbb{P}_P(x < y) \leq 1 - \alpha.$$

The balance constant of P is

$$\delta(P) = \max_{x \neq y} \min\{\mathbb{P}_P(x < y), \mathbb{P}_P(y < x)\}.$$

Theorem 2.3 (The 1/3–2/3 Conjecture). *If P is a finite poset that is not a chain, then $\delta(P) \geq \frac{1}{3}$. Equivalently, P contains a pair (x, y) such that*

$$\frac{1}{3} \leq \mathbb{P}_P(x \prec y) \leq \frac{2}{3}.$$

The upper bound follows immediately from the lower bound $1/3$, since the two probabilities always sum to 1. At first sight the constant $1/3$ may seem mysterious. There is no obvious probabilistic reason that this number should be the correct universal threshold, and the answer to the question was not, at first, very satisfying but the key point is that we cannot ask for anything larger without contradicting some known examples. The simplest one is the disjoint union $C_1 \sqcup C_2$ of a one-element chain and a two-element chain, which can be formulated as such: let

$$P = \{a, b, c\}, \quad b < c,$$

with a incomparable to both b and c .

Example 2.4. *The linear extensions of P are*

$$abc, \quad bac, \quad bca.$$

There are exactly three of them. Therefore

$$\mathbb{P}_P(a \prec b) = \frac{1}{3}, \quad \mathbb{P}_P(b \prec a) = \frac{2}{3}.$$

Similarly,

$$\mathbb{P}_P(a \prec c) = \frac{2}{3}, \quad \mathbb{P}_P(c \prec a) = \frac{1}{3}.$$

It follows that $\delta(P) = \frac{1}{3}$.

If we tried to strengthen the conjecture by replacing $1/3$ with some larger constant $\alpha > 1/3$, then this would be a counterexample. There is another useful way to rewrite the same quantities, and this reformulation is what leads to the counting problem. If x and y are incomparable elements of P , let $P + xy$ denote the poset obtained from P by adjoining the relation $x < y$ and then taking transitive closure. Then every linear extension of P places either x before y or y before x , so

$$e(P) = e(P + xy) + e(P + yx).$$

Moreover,

$$\mathbb{P}_P(x \prec y) = \frac{e(P + xy)}{e(P)}.$$

3 The counting problem

Definition 3.1. *The linear extension counting problem is the function problem that takes as input a finite poset P and returns $e(P)$, the number of linear extensions of P .*

First recall the definition of the class $\#P$.

Definition 3.2. *A function f belongs to $\#P$ if there is a polynomial q and a polynomial-time decidable relation $R(x, w)$ such that for every input x ,*

$$f(x) = \#\{w \in \{0, 1\}^{q(|x|)} : R(x, w) \text{ holds}\}.$$

In words, $f(x)$ counts the number of polynomial-size witnesses w satisfying a condition that can be checked in polynomial time.

For counting problems we usually measure hardness by reductions that are allowed to query an oracle for the target counting problem.

Definition 3.3. *Let f and g be counting problems. One says that f is polynomial-time Turing reducible to g if there is a polynomial-time algorithm that computes f while being allowed to make polynomially many oracle calls to g . A counting problem is $\#P$ -hard if every problem in $\#P$ reduces to it in this sense. It is $\#P$ -complete if it is both $\#P$ -hard and a member of $\#P$.*

The linear extension counting problem is in $\#P$. Indeed, a linear extension of an n -element poset can be encoded by a permutation of its ground set, and we can check in polynomial time whether a given permutation respects all order relations of the poset. The main result we consider is the following theorem of Brightwell and Winkler [2].

Theorem 3.4 (Brightwell–Winkler). *The problem of counting linear extensions of a finite poset is $\#P$ -complete.*

Since $\mathbb{P}_P(x \prec y) = e(P + xy)/e(P)$, exact evaluation of the probabilities in the 1/3–2/3 Conjecture requires exact counts of linear extensions. In particular, deciding whether a pair (x, y) is 1/3-balanced leads naturally to the quantities $e(P)$, $e(P + xy)$, and $e(P + yx)$.

4 Outline of the Brightwell–Winkler proof

In this section we explain the main idea of the proof that counting linear extensions is $\#P$ -complete, following Brightwell and Winkler [2]. For the rest of this section we write $N(P)$ for the number of linear extensions of a poset P , so $N(P) = e(P)$.

The proof is a reduction from 3-SAT Count. What we are trying to show is that given a 3-CNF formula I , we can construct from it posets whose numbers of linear extensions determine the number $s(I)$ of satisfying assignments of I . Once this is done, an oracle for counting linear extensions would allow us to compute $s(I)$, and therefore linear extension counting would be $\#P$ -hard.

The reduction has two main ideas. First, we build a poset $Q_I(p)$ depending on a prime p , in such a way that $N(Q_I(p))$ determines $s(I)$ modulo p . Second, we repeat this for enough primes and reconstruct $s(I)$ by the Chinese remainder theorem.

4.1 3-SAT Count

Definition 4.1. *A literal is either a variable x or its negation $\bar{x}$. A clause is a disjunction of literals. A formula is in 3-conjunctive normal form if it is a conjunction of clauses, each of which contains exactly three literals.*

The problem 3-SAT Count takes as input a formula I in 3-conjunctive normal form and asks for the number $s(I)$ of satisfying assignments of I .

Valiant (here at Harvard!—also the creator of the class $\#P$) proved that 3-SAT Count is $\#P$ -complete [4]. Therefore it is enough to show that, using an oracle for linear extension counting, we can recover $s(I)$ in polynomial time.

4.2 The auxiliary poset P_I

Fix a formula I with m variables and n clauses. Set $M = 7n + m$. Brightwell and Winkler first construct an auxiliary poset P_I with exactly M vertices. For visualizations of the construction, we defer to the original paper on page 235 [2, p. 235].

For each variable x , there is one vertex h_x . For each clause c , say on variables x, y, z , there are seven additional vertices, one above each nonempty subset of $\{h_x, h_y, h_z\}$. Thus each clause contributes seven vertices since for three variables there are exactly seven nonempty subsets.

Write $L_I = N(P_I)$ for the number of linear extensions of P_I . Since P_I has M vertices, any linear extension of P_I is a total ordering of an M -element set, so $L_I \leq M!$.

4.3 Choosing enough primes

Brightwell and Winkler prove a lemma showing that the product of all primes strictly between M and M^2 is at least $M!2^m$ [2]. Since $L_I \leq M!$, the product of the prime divisors of L_I is at most $M!$. Therefore, after removing from that interval all primes dividing L_I , there is still a set S of primes such that

$$\prod_{p \in S} p > 2^m.$$

Since a formula on m variables has at most 2^m assignments, we always have $0 \leq s(I) \leq 2^m$. Hence, if we can determine $s(I)$ modulo every prime $p \in S$, then we can recover $s(I)$ exactly by the Chinese remainder theorem.

Thus the problem reduces to constructing a poset whose number of linear extensions determines $s(I) \pmod{p}$ for each prime $p \in S$.

4.4 The poset $Q_I(p)$

Fix a prime $p \in S$. Brightwell and Winkler construct a poset $Q_I(p)$ of size less than $p(7n + m)$, hence less than M^3 . This keeps the reduction polynomial.

The poset is built around two special vertices a and b . In any linear extension, these split the other vertices into three regions: those below a , those between a and b , and those above b .

Bottom blocks. Below a there is an antichain U of size $(m + 1)(p - 1)$. An antichain means a set of vertices in which no two distinct elements are comparable. The antichain U is partitioned into $m + 1$ blocks of size $p - 1$:

$$U_0, \quad U_x \text{ for each variable } x.$$

Middle blocks. Between a and b there is a second antichain V of size $(n + 1)(p - 1)$, partitioned into

$$V_0, \quad V_c \text{ for each clause } c,$$

again with each block having size $p - 1$.

Literal vertices. For each variable x , there are two vertices labeled x and $\bar{x}$. These are incomparable with both a and b , and each lies above every element of the block U_x .

Clause vertices. Now let c be a clause on variables x, y, z . There are eight possible ways to choose one literal from each of the pairs $\{x, \bar{x}\}$, $\{y, \bar{y}\}$, and $\{z, \bar{z}\}$. So c gives us eight clause vertices $c_1, c_2, \dots, c_8$, one for each such triple. Each clause vertex is placed above the corresponding triple of literal vertices.

Exactly one of these eight clause vertices is placed above b : namely, the one corresponding to the clause c itself. The other seven are placed above every element of the block V_c . In particular, every clause vertex lies above a .

The idea behind the construction is as follows. The blocks U_x , U_0 , V_c , and V_0 are all large antichains of size $p - 1$. If the literal and clause vertices are placed in the wrong way relative to a and b , then

some of these blocks become too isolated, and the number of resulting linear extensions acquires a factor of p . Those contributions therefore vanish modulo p . The only contributions that remain modulo p are the ones encoding satisfying assignments.

It is useful to look at one example. Suppose the clause is $c = (x \vee \bar{y} \vee z)$. Then the eight clause vertices correspond to the eight triples

$$(x, y, z), (x, y, \bar{z}), (x, \bar{y}, z), (x, \bar{y}, \bar{z}), \\ (\bar{x}, y, z), (\bar{x}, y, \bar{z}), (\bar{x}, \bar{y}, z), (\bar{x}, \bar{y}, \bar{z}).$$

The vertex corresponding to $(x, \bar{y}, z)$ is the one forced above b . The other seven are placed above the block V_c .

4.5 Configurations

A linear extension of $Q_I(p)$ determines, for every literal and clause vertex, whether it lies below a , between a and b , or above b . Brightwell and Winkler formalize this by introducing configurations. A configuration is a partition

$$\phi = (B^\phi, M^\phi, T^\phi)$$

of the literal and clause vertices, where B^ϕ is intended to lie below a , M^ϕ between a and b , and T^ϕ above b .

Write L^ϕ for the set of linear extensions of $Q_I(p)$ that realize this placement. If L^ϕ is nonempty, then ϕ is called consistent. In that case L^ϕ is exactly the set of linear extensions of the poset obtained by adjoining the relations $B^\phi < a < M^\phi < b < T^\phi$ and taking transitive closure. Denote that poset by P^ϕ . Then

$$N(Q_I(p)) = \sum_{\phi} N(P^\phi),$$

where the sum runs over all consistent configurations.

4.6 Feasible configurations

Suppose ϕ is consistent. Let P_B^ϕ , P_M^ϕ , and P_T^ϕ be the induced subposets on the bottom, middle, and top sections, respectively. Then

$$N(P^\phi) = N(P_B^\phi) N(P_M^\phi) N(P_T^\phi).$$

A configuration is called feasible if neither $N(P_B^\phi)$ nor $N(P_M^\phi)$ is divisible by p .

Lemma 4.2. *Let R be a finite poset with k vertices, and suppose R has r isolated vertices. Then $N(R)$ is divisible by $k(k-1)\cdots(k-r+1)$.*

Now apply this to the bottom section. The block U_0 always contributes $p-1$ isolated vertices. If neither literal x nor $\bar{x}$ lies below a , then every element of the block U_x is also isolated in the bottom section. The lemma then forces additional divisibility by p . Feasibility therefore forces exactly one of x and $\bar{x}$ to lie below a for each variable x . In other words,

$$B^\phi = \{\text{one literal for each variable}\}.$$

An analogous argument applies to the middle section. The block V_0 contributes $p-1$ isolated vertices, and if no clause vertex attached to c lies in the middle section, then every element of V_c

becomes isolated there. Feasibility therefore forces exactly one clause vertex for each clause to lie between a and b , and no literal vertex may lie there. So

$$M^\phi = \{\text{one clause vertex for each clause}\}.$$

Thus every feasible configuration has the form

$$B^\phi = \{\text{one literal for each variable}\},$$

$$M^\phi = \{\text{one clause vertex for each clause}\}$$

$$T^\phi = \{\text{the remaining literals and clause vertices}\}.$$

4.7 Counting the bottom and middle sections

If ϕ is feasible, then the bottom section consists of the $p - 1$ isolated vertices in U_0 , together with m components, each made of one literal vertex above an antichain of size $p - 1$. They obtain

$$N(P_B^\phi) = \frac{(p(m + 1) - 1)!}{p^m}.$$

To see this, consider a random ordering of the vertices in the bottom section. For a fixed variable x , the probability that its literal vertex lies above all $p - 1$ vertices of the block U_x is $1/p$, since among those p vertices all relative orders are equally likely. These events are independent across variables, so a fraction p^{-m} of all orderings are linear extensions. Since the total number of orderings is $(p(m + 1) - 1)!$, we get the stated formula.

Exactly the same argument yields

$$N(P_M^\phi) = \frac{(p(n + 1) - 1)!}{p^n}.$$

Because $p > m$ and $p > n$, both of these quantities are nonzero modulo p .

4.8 Feasible configurations $\leftrightarrow$ satisfying assignments

If ϕ is feasible, then for each variable x , exactly one of x and $\bar{x}$ lies in the bottom section and the other lies in the top section. Let ℓ_x denote the literal in the top section. Then the set $\{\ell_x : x \text{ a variable}\}$ determines a truth assignment: those literals are declared true.

Now fix a clause c on variables x, y, z . The middle section contains exactly one clause vertex associated with c . Since a clause vertex can lie in the middle section only if all literals below it lie in the bottom section, this middle clause vertex must correspond to the triple of bottom literals. Equivalently, it corresponds to the triple of false literals for that clause.

But one clause vertex was forced above b : the one corresponding to the actual clause c . Therefore the triple of false literals cannot be equal to the clause itself. In other words, the assignment defined by the top literals must satisfy the clause c . Since c was arbitrary, every feasible configuration determines a satisfying assignment of I .

Conversely, suppose we start from a satisfying assignment of I . For each variable x , place the false literal in the bottom section and the true literal in the top section. For each clause c , choose the clause vertex corresponding to the three false literals and place it in the middle section. Since the assignment satisfies c , this vertex is not the one forced above b . Thus this produces a feasible configuration. Therefore feasible configurations are in bijection with satisfying assignments of I .

4.9 The top section

It remains to understand the top section. If ϕ is feasible, then the top literals are the true literals of the corresponding satisfying assignment. For each clause, the one clause vertex removed to the middle section is precisely the one above the three false literals. The remaining seven clause vertices lie above the nonempty subsets of the three top literals. But this is exactly the same pattern used in the definition of the auxiliary poset P_I . Hence for every feasible configuration ϕ , the top section P_T^ϕ is isomorphic to P_I , and therefore

$$N(P_T^\phi) = L_I.$$

Combining the three sections, every feasible configuration contributes the same quantity modulo p :

$$N(P^\phi) = \frac{(p(m+1)-1)!}{p^m} \cdot \frac{(p(n+1)-1)!}{p^n} \cdot L_I.$$

Call this common value N_0 . By the choice of the prime set S , no prime $p \in S$ divides L_I , so $N_0 \not\equiv 0 \pmod{p}$. Since feasible configurations are in bijection with satisfying assignments, we get

$$N(Q_I(p)) \equiv N_0 s(I) \pmod{p}.$$

Thus, once $N(Q_I(p))$ is known, we can recover $s(I) \pmod{p}$.

4.10 The final reconstruction

For each prime $p \in S$, we know $s(I) \pmod{p}$. Since the primes in S are distinct, they are pairwise coprime, so by the Chinese remainder theorem (which is not referred to by name in the paper which made looking up and understanding their use of it much harder) these residues determine $s(I)$ uniquely modulo $\prod_{p \in S} p$. Because

$$0 \leq s(I) \leq 2^m \quad \text{and} \quad \prod_{p \in S} p > 2^m,$$

this determines $s(I)$ uniquely as an integer.

Therefore an oracle for counting linear extensions computes the number of satisfying assignments of a 3-CNF formula in polynomial time. This proves that linear extension counting is $\#P$ -hard. Since it also belongs to $\#P$, it is $\#P$ -complete.

Theorem 4.3. *The problem of evaluating $\mathbb{P}_P(x \prec y)$ exactly is polynomially equivalent to the problem of counting linear extensions.*

The forward direction is immediate from $\mathbb{P}_P(x \prec y) = e(P + xy)/e(P)$. Conversely, Brightwell and Winkler show that one can recover $e(P)$ from a polynomial number of exact probability evaluations by imposing relations one at a time along a comparison-sorting process, and we defer to their paper for the proof for fear of not having enough time to study for a final in 2 days.

References

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